INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

CLEANER, STEAM, PRESSURE JET, DIESEL FUEL FIRED, SKID MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a skid mounted, electric motor driven, diesel fuel fired, pressure jet steam cleaner and pressure washer. The cleaner specified is the continuous coil type, with no space for steam accumulation, in which water is heated to a temperature below the saturation temperature for the pressure in the coil and is discharged from the nozzle at high velocity. The cleaner should discharge no less than 600 gallons per hour (gph) of saturated steam at a pressure of no less than 200 pounds per square inch gage (psig) and no less than 500 gph of hot and cold water at a pressure of no less than 400 psig.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation (see 6.2.1).

SPECIFICATIONS

FEDERAL

A-A-870	- Antifreeze/Coolant Engine: Ethylene Glycol, Inhibited Concentrated
A-A-1898	- Cushioning Material, Cellulosic, Packaging
L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
P-S-120	- Scale Removing Compound
TT-P-664	- Primer Coating, Alkyd, Corrosion-Inhibiting, Lead and Chromate Free, VOC-Compliant
ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water Suction and Discharge

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940

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ZZ-H-601 - Hose and Hose Assemblies, Rubber (Wrapped or Braided) Water Service
PPP-B-636 - Box, Shipping, Fiberboard
PPP-B-1055 - Barrier Material, Waterproofed, Flexible
PPP-C-850 - Cushioning Material, Polystyrene, Expanded, Resilient (for Packaging Uses).
PPP-C-1797 - Cushioning Material, Resilient, Low Density, Unicellular Polypropylene Foam
PPP-T-60 - Tape, Packaging, Waterproof

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproof, Waterproofed, Flexible
MIL-V-173 - Varnish, Moisture and Fungus Resistant (For Treatment of Communication, Electronic and Associated Equipment
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for
MIL-E-16298 - Electric Machines Having Rotating Parts, Accessories and Associated Support Items: Packaging Of
MIL-B-22191 - Barrier Material, Transparent, Flexible, Heat Sealable
MIL-C-22542 - Cleaning Compound, High Pressure Cleaner, Liquid

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards For Federal Services
FED-STD-101 - Test Procedures for Packaging Materials

MILITARY

MIL-STD-129 - Marking For Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Attachment Office, Bldg 4D ATTN: NPM-DODSSP, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the

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issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2.1).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI S1.4 - Sound Level Meters

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A178 - Electric-Resistance-Welded Carbon Steel and Carbon-Manganese Steel Boiler Tubes
ASTM A214 - Electric-Resistant-Welded Carbon Steel Heat-Exchanger and Condenser Tubes
ASTM A513 - Electric-Resistance-Welded Carbon and Alloy Steel Mechanical Tubing
ASTM D2156 - Smoke Density in Fuel Gases from Burning Distillate Fuels

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02669.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street NW, Washington, D.C. 20037.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J356 - Welded Flash Controlled Low Carbon Steel Tubing Normalized for Bending, Double Flaring and Beading

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

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(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall furnish one or more complete steam cleaner(s) for First Article Inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design.

3.2.1 Design construction. The cleaner shall be designed for one gun and two gun operation and shall provide both steam and water (hot and cold) sprays to remove dirt, grease, and heavy oil deposits. The cleaner shall be designed for use with liquid self-emulsion cleaning compound conforming to MIL-C-22542 and for operation with fuels in the classification of diesel fuel grades Number 1 and Number 2. The unit shall have an output capacity of 600 gph in the steam vapor discharge mode and 500 gph in the hot and cold water discharge modes. Components shall be applied in accordance with standard commercial practice and shall function within rated design capacity. Components of the cleaners shall be designed, located, constructed or protected so that exposure to the weather shall not interfere with any operation or function specified herein. Components shall be mounted on a common chassis of welded steel construction. The cleaner shall be designed to withstand multiple relocations without damage and without the need for any disassembly. Drive chains and variable pulleys shall not be used. Mounted motors and pumps shall not be operated above the manufacturer's recommended rated revolutions per minute (rpm) for the purpose of meeting the performance requirements specified herein. The steam cleaners shall perform as specified when operating from a 208 volt $\pm$ 10 percent, single phase, 50/60 Hertz (Hz) power supply. The power supply may be transformed at the control panel to operate control circuits at lower voltages.

3.2.2 Materials. Materials not specifically designated herein shall be at least equal to those being supplied in the manufacturer's commercial steam cleaners for which a proven service record exists and which have been accepted by industry for the use intended. Where dissimilar metals are in contact, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

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3.2.3 Workmanship. The quality of workmanship imparted to the item and all of its components shall equal or exceed that typically provided to commercial products by domestic producers of the type of item addressed herein. The items presented for acceptance shall have been manufactured with skill and care; shall be uniform, neat, and clean; and shall be free of irregularities and anomalies which degrade form, fit, function, performance or appearance.

3.3 Performance and product characteristics.

3.3.1 Discharge requirements. The cleaner shall be provided with stop and start controls as well as regulating controls to achieve and automatically maintain a burner capacity commensurate with any desired vapor or pressure rinse output. The steam cleaner shall meet the pressure and discharge requirements of this specification when operating with either one or both guns. The two-gun discharge pressure shall be within 25 psig of the one-gun discharge pressure, but still no less than that required in 3.3.1.1.

3.3.1.1 Vapor discharge. The cleaner, with or without cleaning compound, shall have the ability to raise and maintain the temperature of no less than 600 gph of no greater than 40 degrees Fahrenheit (F) temperature inlet water to an outlet temperature of 300 to 350 degrees F within eight minutes of initial combustion from a cold start and discharge it from a spray gun at a discharge manifold pressure of no less than 200 psig. The cleaner shall be suited for one and two gun operation as follows:

a. No less than 600 gph at no less than 200 psig through one gun.

b. No less than 300 gph at no less than 200 psig through one gun with the appropriate smaller nozzle.

c. No less than 300 gph at no less than 200 psig through each of two guns with no greater than 25 psi difference between the two guns.

3.3.1.2 Vapor discharge continuity. The cleaner shall allow switching from two gun operation at 300 gph per gun to one gun operation at 300 gph without stopping the cleaner.

3.3.1.3 Hot water discharge. The cleaner shall have the ability to raise and maintain the temperature of no less than 500 gph of inlet water (at no greater than 40 degrees F) to an outlet temperature of not less than 160 degrees F and discharge it through the rinse gun at no less than 400 psig as measured at the discharge manifold.

3.3.1.4 Cold water discharge. The cleaner shall have the ability to discharge no less than 500 gph of inlet water (at no greater than 50 degrees F) through the rinse gun at a pressure of no less than 400 psig as measured at the discharge manifold.

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3.3.1.5 Thermal efficiency. In both the hot water and steam modes, the cleaner shall have a thermal efficiency no less than 80 percent while operating at full capacity. The efficiency shall be met using No. 1 and 2 diesel fuel and on 50 and 60 Hz power sources. Tests may be conducted with No. 2 diesel fuel only. The thermal efficiency after 200 hours of operation shall be no less than 80 percent. Thermal efficiency is defined as the ratio of the heat energy units added to the discharged water divided by the heat energy units released by the oxidation of the fuel. Stack temperatures measured no more than six inches from the exhaust opening shall not exceed 450 degrees F above ambient temperature. Smoke density measured in accordance with ASTM D2156 shall be no greater than No. 2 during the first 400 hours of operation. In order to achieve the efficiencies required, a heat reflector shield may be installed between the outside of the coil and any insulation to allow heat flow around the coil periphery. A cold water coil may also be used outside the hot water coil to achieve the required thermal efficiencies. Readjustment of fuel delivery rate is not permitted to satisfy other performance requirements.

3.3.2 Continuous operation. The cleaner shall withstand without damage, and shall perform within the requirements of 3.3.1, while being subjected to continuous operation as follows:

a. As a vapor cleaner, for a period of 400 hours.

b. As a hot rinse unit, for a period of 48 hours.

The steam cleaner shall be capable of operating for no less than the periods stated above without requiring desooting. Smoke density, measured as in ASTM D2156, shall be no greater than number 2 at the beginning and throughout the above stated periods. Steam cleaners shall also be capable of operating no less than the periods stated above with no breakdown or replacement of major components, that is, pumps, motors, burners, safety controls and devices, heating coils, or transformers. Compliance with this continuous 400 hour steam mode/vapor discharge and the 48 hour hot water rinse test will be performed for no less than 8 hours per calendar day and shall be completed within 56 consecutive calendar days.

3.3.3 Tilted, position operation. The cleaner shall not leak fuel and shall perform within the requirements of 3.3.1 when all the tanks are filled to their full mark and the cleaner is tilted to any and all angles from a horizontal plane to 15 degrees in any and all directions from the horizontal plane.

3.3.4 Altitudes. The cleaner shall perform within the requirements of 3.3.1 at altitudes up to 6,000 feet in ambient air temperatures up to plus 107 degrees F. Stoichiometric calculations based on fuel analysis and velocity pressure in the combustion chamber may be used in lieu of high altitude tests.

3.3.5 Operating temperatures. The cleaner shall meet the requirements of 3.3.1 while being operated in air temperatures from plus 35, -0 +5, degrees F to 125, -5 +0, degrees F.

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3.3.6 Storage temperatures. The steam cleaner and all components shall withstand without damage, and shall meet the requirements of 3.3.1, when stabilized at a temperature of plus 80, ±5, degrees F after having been subjected to:

a. A temperature of minus 50, ±5, degrees F for a period of 72 hours.

b. A 24 hour cycle of temperatures, starting with ten hours at plus 80 degrees F, followed by five hours of steady increase to plus 150 degrees, ±5, F, four hours at plus 150 degrees F and five hours of steady decrease to plus 80 ±5, degrees F.

3.3.7 Transport. The design of the steam cleaner shall incorporate the capabilities to accommodate without degradation (cracked welds, permanent deformation, damaged equipment) the following:

a. Truck and trailer bed transport at speeds up to 50 miles per hour (mph) on primary roads, up to 35 mph on secondary roads, and up to 5 mph on rough terrain.

b. Truck and trailer bed operation.

c. Fork lift handling.

d. Hoisting and lifting equipments (cranes, hoists, helicopters, etc.).

3.3.8 General construction. All parts and materials used in fabrication of the cleaner shall be new and free of defects. In no event shall patching processes be used for reclaiming defective parts. All surfaces of all parts shall be clean and free of extraneous materials. External surfaces shall be free of sharp edges and burrs. All screws, pins, and similar parts shall be installed to prevent loss of tightness. However, such parts subject to adjustment or removal shall not be permanently deformed.

3.3.8.1 Electrical construction. All controls and wiring shall conform to NFPA 70 and shall be designed, located, constructed or protected so that exposure to the weather shall not interfere with any operation specified herein. Connections of conductors and terminal parts shall be of the screw, pressure, or solder type. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Only resin base fluxes shall be used in soldering operations. Connections to screw type terminals shall be mechanically secured to prevent loss of tightness. All wiring shall be secured and protected to prevent contact with hot surfaces, vibration, and chafing. There shall be no exposed electrical connections.

3.3.8.2 Service accessibility. The cleaner shall be constructed to provide ready access to components so that maintenance can be performed using only normally available standard tools and maintenance equipment. The cleaner shall be designed and constructed so that any repair, including coil replacement, can be accomplished in no more than 4 hours by one person.

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3.4 Maintainability. Steam cleaners using water with a hardness of no less than 12 grains per gallon shall be capable of:

a. Performing to full capacity in the steam mode no less than 8 consecutive hours without requiring blow down.

b. Performing to full capacity in the steam mode no less than 80 hours without requiring chemical descaling.

c. Performing to full capacity in the steam mode no less than 400 consecutive hours without desooting.

d. Performing in the steam mode for 400 hours and in the hot water rinse mode for 48 hours.

e. Performing each 40 hours, at full steam capacity, with no more than 5 hours of preventative maintenance. Preventative maintenance includes blowdown but does not include chemical descaling.

f. Operating for 448 hours with no breakdown or replacement of major components, that is, pumps, motors, burners, safety controls and devices, heating coils, or transformers. The maintainability test shall be completed within 56 calendar days.

Cleaners shall have the capability of a forward blowdown cycle. The blowdown removes scale and deposits and increases the time between descalings by allowing the heat to remain on for a time with no water supply to the coil. The heat dries and cracks the deposits. A full volume of water is then introduced into the hot coil. This water flashes to steam and creates sufficient pressure to loosen scale and sludge and flush it out of the coil. Blowdown controls and steam outlets shall be designed and located to preclude operator injury and steam cleaner damage. Blowdown instructions and safety notes shall be permanently attached to the steam cleaner near to and readable at the blowdown controls. Any tools or equipment other than common hand tools required to adjust the burner or maintain the cleaner shall be furnished with each cleaner. All lubrication points shall be accessible without the need for disassembly of the steam cleaner. All vent openings shall be temporarily closed to prevent loss of lubricants during shipment. Warning tags shall be attached and the supplier shall include a warning notice in the instruction manual, to remove the temporary caps and stoppers, to prevent the start-up or use of the steam cleaner without removal of the temporary caps and stoppers. Compliance with this 400 hour steam mode/vapor discharge and the 48 hour hot water rinse test will be performed and shall be completed within 56 consecutive calendar days. During the 400 hour steam mode/vapor discharge test and the 48 hour hot water rinse test the gun(s) valve(s) shall be closed at 1 hour +15 minute intervals so that water flow from the nozzle(s) is(are) stopped and fuel flow to the burner stops. At least every 4 hours of operation the cleaner shall be stopped, allowed to cool and restarted.

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3.5 Safety provisions. Protection against electrical, mechanical and thermal hazards shall be incorporated into the cleaner. Steam and water outlet fittings and safety relief unloader valves shall be located away from electrical controls and the operator's control panel. The operator, in the normal operation of the steam cleaner, shall be protected from contacting any component surfaces that exceed 130 degrees F by positioning relative to cooler components, by direct shielding, or by insulation. Typical components are: burner assembly, coil assembly, coil jacket, and vapor/hot water piping. Valves or controls that operate at temperatures greater than 130 degrees F shall be insulated so that contact surface temperatures do not exceed 130 degrees F. There shall be no fuel leakage during any operational or idle phase. V-belt drives, if used, shall have protective safety shielding.

3.5.1 Power input. A manually actuated master switch shall provide on/off for all electrical power to, and used by, the cleaner. The master switch shall not allow the cleaner to restart automatically after a power failure or interruption.

3.5.2 Burner operation. Steam cleaners shall have safety devices that will terminate burner operation by shuting off, at a minimum, the fuel flow to the burner when:

a. Flame control. There is no flame in the burner combustion chamber. The flame control sensing device shall be either flame or heat sensing and shall not be aided by reflective devices. A spark sensing type shall not be permitted. Manual reset shall be required for reignition. Operation shall be terminated for:

(1) A flame is not established within 45 seconds during starting.

(2) A flameout occurs.

b. Inlet water flow control. The water flow to the pump is interrupted or reduced to less than 150 gph due to any of the following:

(1) Interruption of flow in the pressure water supply to supply tank.

(2) Blockage and/or collapse of the suction line.

(3) Restriction in the inlet strainer.

c. Outlet water flow control. The water flow from the pump is blocked or reduced to less than 150 gph due to any of the following:

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(1) Coil inlet downstream of the pump is blocked.

(2) Increased resistance to water flow through the coil.

(3) Coil outlet is blocked.

(4) Delivery gun valve is closed.

d. High temperature limit device. The fluid temperature reaches 360 degrees F +0 -25 degrees F in the vapor mode and 260 degrees F +0 -25 in the hot water mode. Water shall continue to flow through the coil.

e. High pressure limit device. The fluid pressure at the coil outlet approaches 600 psig +0 -25 psig in the hot water mode and 400 psig +0 -25 psig in the steam mode. Water shall continue to flow through the coil.

For conditions b through e the burner shall automatically reignite when the stated condition is corrected. When changing from steam vapor mode to high pressure rinse mode, the high temperature and high pressure limiting devices shall be changed to the correct limiting value automatically. Because of the potential hazard to personnel if steam pressures exceed the steam hose rated working pressures, controls design shall prevent temperatures from exceeding 210 degrees F when the pressure by-pass system is in the high pressure rinse mode and the temperature or mode selector is in the steam mode. When the cleaner is operating in the steam mode, the output pressure and temperature shall not exceed the rated working pressure and temperature of the steam hose.

3.5.3 Bypass system. In addition to the above, each steam cleaner shall have a mechanical or hydraulic (non-electric) bypass system that will bypass the entire pump output to the pump inlet when:

a. The working pressure is exceeded for the mode of operation. Bypass must actuate within -0, +175 psig of the specified working pressure.

b. The coil inlet/coil outlet is blocked.

c. The delivery gun valve is closed.

During bypass the pump motor shall keep running. The pressure regulator devices for each mode of operation shall be separate and shall be automatically engaged. Readjustment of a regulator shall not be required when changing from the low pressure vapor mode to the high pressure rinse mode.

3.5.4 Safety relief system. A self-resetting pressure relief valve shall be mounted at the coil outlet. The relief valve shall have the capacity to relieve the full pump output pressure. The valve shall discharge to atmosphere toward the ground and away from the operator. The pressure relief valve shall discharge the fluid upon reaching 700 to 800 psig.

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In testing for article 3.5 safety provisions, the high temperature and high pressure limit devices, as well as the bypass and safety relief system, shall be disconnected or defeated as necessary to complete the tests.

3.5.5 Electrical. The conductors and all electrical components shall withstand, without damage, twice the rated voltage plus 1000 volts root mean square (rms) at 60 Hz for a period of 60 seconds between mutually insulated circuits, and between ground and insulated circuits. All wires, conduit, and electrical components shall withstand the temperatures of normal operation. All wiring shall conform to NFPA 70. All wiring and conduit shall be secured to prevent chafing and contact with hot components. Provision shall be made to assemble and wire all components in a way that assures equipment grounding in accordance with NFPA 70, Section 422-8. There shall be no exposed electrical connections.

3.5.6 Noise level. The sound level of the cleaner under all operating modes, excluding the discharge gun(s) sound, shall not exceed 85 decibels (dbA). Measurements are to be made on the A scale at slow response on a sound level meter meeting the Type I requirements of ANSI S1.4. Readings shall be taken at a height of five feet above the ground plane and at ten feet from the perimeter of the cleaner. Apply Stocher decal #1453 (International Hearing Protection Symbol) less "Danger" in a visible area as an operator warning.

3.6 Water supply system. The cleaner shall be equipped to secure water from the following sources:

a. Tap pressure system from a minimum pressure range of 25 to 125 psig.

b. Suction line or vacuum pickup from well, stream, or other outside source of water.

3.6.1 Water supply tank. A water supply tank shall be furnished to provide a constant supply of water to the cleaner (from the source specified in 3.6 (a)) to meet all performance requirements specified herein. The water tank shall be stainless steel conforming to one of the following AISI designations: 302, 304, 316, 317, 318, 319, 347, or 348. Surface finish is optional. The tank shall be located for easy accessibility and shall be such that leakage, spillage, splashing or overflow of water shall not come in direct contact with electrical components. The tank shall be provided with a drain valve, piped to the cleaner exterior, which shall permit complete draining of the tank. The tank outlet to the pump shall have a fine mesh stainless steel screen. The tank shall be designed and manufactured to prevent back siphonage into the water supply in the event of reduced pressure or failure of the water supply.

3.6.2 Water supply hose and connections. The water hose shall conform to grade 1 of ZZ-H-601. It shall be a minimum of one inch inside diameter and fitted with standard male and female hose couplings with 0.75-11.5 National Hose coupling threads. One water supply hose having a minimum length of 50 feet shall be supplied with each cleaner.

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3.6.3 Water supply suction hose. The cleaner shall be supplied with a suction hose (in addition to the hose in 3.6.2) to permit the cleaner to obtain water from a well, stream, or other outside source. The suction hose shall be a minimum of 25 feet in length. It shall have an inside diameter of 1.25 inches, shall conform to grade B, type I of ZZ-H-561, and shall be provided with all necessary standard couplings. A foot valve and strainer shall be provided for attachment to the suction hose. The strainer shall be designed to keep the foot valve suspended so that the valve does not become clogged or bury itself in the bottom of the tank or well.

3.6.4 Pump. The pump shall be self-priming (a quick air release may be incorporated to comply with the self-priming criteria if necessary), shall be corrosion resistant to water, cleaning solutions, and acid deliming solutions, and shall be of sufficient capacity to lift and deliver water to meet the requirements of 3.3.1 to 3.3.2. The pump shall not be damaged in the deliming process. A Y-pattern or T-pattern sediment strainer, with a full flow blow-off valve on the branch, shall be provided before (ahead of) the pump inlet. The strainer shall contain a corrosion resistant screen adequate to protect the water pump from injurious contaminants but with no greater than .02 inch openings. The pump output shall be by-passed anytime there is a flow restriction that could damage the pump. Reciprocating or impulse-type pumps shall be equipped with a shock absorbing device; e.g., pulsation dampener in the discharge line. The pump shall be equipped with connections to allow pumping water from a tank, stream, or outside source directly through the coil and dispensing system. The pump shall be capable of lifting water, to meet the requirements of 3.3.1 to 3.3.1.3, no less than 20 feet at sea level. The pump shall show no signs of wear or damage from repeated starts and stops when obtaining water from a tank or well. Provision for draining the pump shall be incorporated.

3.7 Heating coil. The heating coil shall be manufactured from round, low carbon, steel tubing suitable for bending. The tubing used to form the coil shall be one continuous length without any transverse or repair welds. Electric-resistance-welded steel tubing shall have a minimum inside weld flash and be free of splatter, etc. to reduce localized discontinuities which are subject to concentration of corrosion and lime deposits. In all cases, the inside longitudinal weld flash height shall not be greater than the wall thickness or .094 inch whichever is less. The seam weld shall be nondestructive tested to identify and reject tubes containing injurious defects. The tubing shall be annealed or normalized to improve bending. Heat treatment to 1650 degrees F of the seam weld only is acceptable on electric-resistance-welded tubing. A flattening test on a shipment test specimen shall be performed according to applicable ASTM or SAE standards as an indication of bending capability. All tubing shall withstand a hydrostatic pressure proof test, with no evidence of failure, at a fiber stress of 80 percent of the material minimum yield. Hydrostatic test pressure shall be no less than 2000 psi, but need not be tested beyond 5000 psi. Test pressure is to be calculated by Barlow's formula for thin hollow cylinders under tension:

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$$P = 2St/D$$

Where: P = hydrostatic test pressure, psi
S = allowable fiber stress (80 percent of min yield strength) lbs/in
t = specified minimum wall thickness, inch
D = specified outside diameter, inch

The tubing producer or supplier shall furnish to the steam cleaner manufacturer a certificate of compliance to these performance requirements. Note: ASTM A513, ASTM A214, ASTM A178 and SAE J356 tubing will meet these requirements.

There shall be no circumferential welds or joints in the coil except those required to attach end fittings. The completed coil assembly shall withstand a hydrostatic pressure test of 2000 psi for no less than 10 minutes with pressure drop not to exceed 10 psig.

Following the 2000 psi hydrostatic testing and before installing in the cleaner the completed coil assembly shall be stress relieved by heating to 1100 to 1200 degrees F. The entire assembled water piping system shall withstand a hydrostatic pressure of 1.5 times the maximum relief valve pressure (1050 psig) for no less than ten minutes with a pressure drop not to exceed 10 psig. The coil shall be designed for easy removal and replacement without damage to the coil, coil jacket, or supports using only common hand tools (pipe wrench, end wrench, etc.). For delivery, the vapor and water system shall be protected against corrosion as specified in Section 5 of this specification or elsewhere in the contract.

3.7.1 Coil pressure gages. Dial-type pressure gages, having a range not less than 0 to 600 psig, shall be attached at the intake and outlet of the heating coil. The gages shall be damped to permit clear readings. In addition, each cleaner shall have a temperature gage to monitor the outlet temperature. All gages shall be no less than three inches in diameter and shall be mounted on the control panel. All gages shall withstand the temperatures and pressures of blowdown without damage, or the pressure gages shall be equipped with shut-off valves, accessible from the control panel, to isolate them during blowdown. Gages shall be accurate within five percent of full scale value.

3.7.2 Coil valves and fittings. Suitable valves and fittings shall be provided on both the intake and outlet of the heating coil to permit complete drainage and blowdown and to permit the introduction and circulation of a scale removing compound conforming to P-S-120. The solution acts as a scale remover without the need for an auxiliary pumping medium. All necessary hoses, screening, and filtering required to perform scale removing, as recommended by the manufacturer, shall be provided with each cleaner.

3.7.3 Blowdown provisions. The steam cleaner shall have provisions for periodic blowdown for removing scale and other deposits from the inside of the coil to minimize the necessity for frequent descaling. The blowdown procedure shall consist of allowing the fire to remain on, with no water supply to the coils, for the length of time specified by the manufacturer so that the heat

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dries and cracks the deposits. The pump shall continue running with pump output bypassed or, if an electromagnetic clutch is used, the drive motor shall continue to operate. Then allow a full volume of water to enter the hot coil where it flashes into steam and creates sufficient pressure to loosen scale and sludge and blow it out of the coil. Blowdown controls and steam outlets shall be designed to preclude injury to the operator and damage to the steam cleaner. Blowdown instructions and safety caution notes shall be illustrated and permanently attached to the steam cleaner near the controls. The steam cleaner shall be capable of operating in the vapor mode no less than 8 consecutive hours without requiring blowdown and no less than 80 consecutive hours without chemical descaling when using water with hardness no less than 12 grains per gallon.

3.8 Burner. The burner shall be of the fuel pressure atomizing or air atomizing type equipped with an ignition system using a 50/60 Hz electric spark. Provisions shall be incorporated to prevent any fuel leakage into the combustion chamber until proper fuel pressure is obtained. All wiring and controls for the ignition system shall be designed, located, constructed or protected so that exposure to the weather shall not interfere with any operation specified herein. The wiring shall conform to NFPA 70.

3.8.1 Burner position. The burner shall be positioned to facilitate cleaning and replacement and shall be designed or located to prevent condensation from damaging the burner components. The burner shall be so designed that removal and replacement can be accomplished without lifting any portion of the cleaner from the ground or removing the cleaner from the bed of the trailer, if so mounted. Operation of the burner shall not be affected by vibration or sudden shock.

3.8.2 Combustion chamber. Deflectors and the top of the combustion chamber shall be corrosion resistant.

3.9 Fuel tank. The fuel tank shall be fabricated of standard commercial mild steel. For delivery, the tank shall be protected against corrosion as specified in Section 5 or elsewhere in the contract. The fuel tank capacity shall permit no less than four hours of continuous operation under maximum heat load. The tank shall be subject to atmospheric pressure only and shall be protected to prevent possible seepage, splashing, or leakage of other liquids into the tank. The tank filler opening shall have a minimum internal diameter of 2.0 inches, a removable 20 mesh (.033 inch opening) strainer, and a chain-secured, vented filler cap. The fuel tank shall be located to prevent fuel from dripping onto any component of the electrical system. A shutoff valve and a dirt and water separator with a removable 100 mesh (.005 inch opening) fuel strainer or a fuel filter shall be installed at the fuel tank outlet. In addition, a drain valve shall be incorporated with such standard pipe and fittings as are required to provide accessibility for manual control of discharge flow. The drain outlet shall be located so that the fuel cannot drain onto other components or structural members of the cleaner. The fuel tank shall incorporate a break resistant visual gage that allows the fuel level to be observed while filling the tank. A dip stick shall not be acceptable. The visual gage shall be provided with protective guard(s) to

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prevent breakage. The fuel lines shall be copper, brass, or steel tubing or pipe. Fittings, other than steel pipe fittings, shall be the flare type. All fuel lines shall be anchored to prevent movement, vibration and chafing.

3.10 Cleaning compound system. The cleaner shall be provided with at least one cleaning compound tank, of no less than 10 gallon capacity, fabricated of either stainless steel of the same material as specified in 3.6.1 or high density polyethylene no less than 0.100 inch thick. The cleaning compound tank shall be equipped with means for metering and introducing the liquid compound into the water flow at various rates. Provisions shall be made to prevent back siphonage of the cleaning compound into the water supply. The tank shall be equipped with a compound level indicator marked in intervals no greater than one gallon. The level indicator shall indicate the level in the tank while the tank is being filled. If the tank is transparent or translucent, no level indicator is needed. The level indicator shall be protected against breakage and shall not be made of glass. The cleaning compound tank shall be vented and shall have a filler opening of no less than 2 inches inside diameter. The filler cap shall be chained or cabled to the tank or cleaner frame. The cleaning compound tank shall be equipped with a drain.

3.11 Control panel. The steam cleaners shall have a control panel mounted in a location convenient to the operator during operation, but such that the operator is not near the discharge opening. The control panel shall contain, but not be limited to: a coil inlet pressure gage, a coil outlet pressure gage, a fuel pressure gage, the cleaning compound metering valve, a master power switch of the circuit breaker type, a burner switch, a pump switch, a coil outlet temperature gage, a cleaning mode switch, the blow down switch, an hour meter, and any other gages, switches, or controls required for operation. All gages, switches, and controls shall have their function plainly marked. Gages shall be no less than 3 inches in diameter and accurate to within 5 percent of full scale. The fuel pressure gage shall have a range of no less than 0 to 300 psi and shall be dampened against pressure pulsations and fluctuations. All other gages shall have a range of no less than 0 to 600 psi and shall be dampened against pressure pulsations and fluctuations. Pressure gages shall withstand or be protected against the rigors of repeated blow down. Valves that isolate gages during blow down shall be accessible from the control panel. The hour meter shall register to 9999.9 hours in 1/10 hour increments and shall not be able to be reset.

3.12 Skid mount. The cleaner shall be equipped with a welded, heavy structural steel frame with two runners. All frame members shall be braced and gusseted to provide a rigid, sturdy base and to prevent sagging or distortion of the frame. The runners shall be curved upward at both ends to facilitate pulling the cleaner over rough or uneven ground. Both ends of each runner shall be equipped with eyes, not less than three inches inside diameter, to allow the cleaner to be lifted and towed. The skid shall allow for operation and maintenance of the cleaner without removal. Skid runners shall be designed so that water, fuel, or debris shall not collect and stand. Pipe or tube runners shall be capped on both ends to prevent water, fuel, and debris from entering. The cleaner frame and skids shall be designed so that

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the cleaner, with maximum load, can be lifted by standard fork lift from each of the four sides without damage. The use of fork extensions is not permitted.

3.13 Delivery equipment. Delivery equipment shall consist of delivery hose, guns, nozzles, fittings, valves, and drain valves necessary for safe and efficient operation of the cleaner. An outlet manifold with shutoff valves shall be provided at the cleaner discharge for controlling the operation of each delivery gun. The cleaner shall have sufficient discharge outlets to permit performance of the vapor spray and rinse operations specified in 3.3.1.1 through 3.3.1.3 without relocation of any hose. The manifold outlets shall be equipped with swivel couplings for the attachment of the hoses.

3.13.1 Delivery hose. The hoses shall be rated for a working steam pressure of no less than 250 psig and a temperature of 450 degrees F. The hoses shall also be rated for high pressure and hot water applications of no less than 600 psig and 250 degrees F. The burst pressure shall be no less than 2500 psig. The hose shall have an inside diameter of no less than 3/4 inches. Standard male pipe thread hose nipples shall be held in place with two bolt interlocking type clamps. The cleaner shall be supplied with a sufficient number of 50 foot hoses to perform the operations specified in 3.3.1.1 through 3.3.1.3 without relocation of any hose. Hose shall be braided steel reinforced with a protective cover that is oil and abrasion resistant with at least 50 effective pricked holes per foot.

3.13.2 Delivery guns. The cleaning gun assembly shall not be less than 4.5 feet in overall length and shall be equipped with a self-closing shut-off valve to control steam flow at the nozzle. The discharge end of the wand shall be straight and have a removable 45 degree fitting threaded such that the delivery nozzles can be mounted on either the straight wand or the 45 degree fitting. The cleaner shall permit remote control by the use of this valve. Discharge from the cleaner shall not occur until the valve is opened by the operator. The valve shall be located on or near the rear handle grip and its use concurrent with the operator's secure hold on the assembly. The valve shall shut off when the operator removes his hand from the rear grip of the gun assembly. Disassembly of the valve for removal of scale particles in the water passages shall be possible using standard tools. The forward gun assembly shall include a cylindrically shaped front swivel handgrip, no less than 12 inches long, insulated and shielded to protect the operator from heat when handling without gloves. The swivel assembly shall rotate the entire portion of the cleaning gun forward, and inclusive of, the front handgrip 360 degrees. The cleaner shall be provided with two cleaning gun assemblies to perform the operations specified in 3.3.1.1 through 3.3.1.3.

3.13.3 Delivery nozzles. Nozzles shall be designed and fabricated for corrosion and wear resistance. The following nozzles shall be furnished to operate the cleaner at rated pressure (see 3.3.1.1 and 3.3.1.2):

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a. One straight, round high-pressure vapor nozzle, 600 gph minimum.

b. Two straight, round high-pressure vapor nozzles, 300 gph minimum.

c. Four 30 degree offset flat fan shape, high-pressure vapor nozzles, 300 gph minimum.

d. Two 30 degree offset flat fan shape, high-pressure vapor nozzles, 600 gph minimum.

e. Two straight high-pressure rinse nozzles, 500 gph minimum.

3.14 Instructions. Instructions on the use, repair, and maintenance of the cleaner shall be included with each cleaner. In addition, each cleaner shall have an instruction panel attached near and readable from the control panel containing:

a. Starting and stopping procedures.

b. Operating instructions, including detailed instructions for obtaining the proper percentage of cleaning compound and the blowdown procedure.

c. The words, "Frequency of maintenance to be as specified in the instruction manual." Alternate method, specify maintenance requirements by frequency on the instruction plate (i.e. all daily, monthly, etc.), list as a separate entry.

The words, "Use only Diesel fuel grades Numbers 1 or 2" shall be permanently and legibly marked in letters no less than 1 inch high on the outside front of the cleaner. The instruction plate shall be impervious to oil, grease, cleaning compound, water, and weather. The instructions shall be etched or embossed plastic or metal or an adhesive backed metal (mylar) type plate. Decals are not acceptable. All valves, gages, controls, and indicators used in the operation of the shipment shall be identified for function and direction of operation by securely attached nameplates of the same type stated above. Tags are not acceptable. The cleaner shall have an identification plate as above with the following information:

NOMENCLATURE: CLEANER, STEAM PRESSURE JET, SKID MOUNTED
NSN:
SERIAL NUMBER:
CONTRACT NUMBER:
MFR NAME AND ADDRESS:
VOLTS:
HERTZ:
PHASE:
AMPS:

Where practical, all components and subassemblies shall be marked with the manufacturer's or vendor's part number.

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3.14.1 Starting sequence. Water flow and combustion air flow shall be established and maintained before fuel is supplied to the burner nozzle.

3.15 Electric motors. The drive motor shall be an induction, squirrel cage type conforming to NEMA MG-1. The motor shall operate on 208 volt ± 10 percent, 50/60 Hz, single phase alternating current, shall have sealed ball or roller bearings, shall be drip proof or totally-enclosed fan cooled, and shall be designed for continuous duty operation. The motor shall have Class B insulation or better. The motor shall be provided with integral manual reset, thermal overload protections. Auxiliary motors shall be standard commercial types, conforming to MG-1 as applicable. All motors shall be provided with the motor manufacturer's standard commercial nonhygroscopic treatment compatible with steam service. As an alternate to the motor integral manual reset, a full voltage starter with manual reset for squirrel cage motors may be used to provide both running thermal overload and locked rotor thermal overload protection.

3.16 Power cable. The cleaner shall be supplied with not less than 75 usable feet of No. 14 gage (or larger, as required by current draw) three-conductor cable. The cable shall be Underwriters Laboratories Type SO, or equal. A flexible strain relief connector shall be attached to the cleaner and the power cable in such a manner as to prevent tension or strain at the cable electrical connections. The cable conductors and ground shall terminate in one inch of stripped, tinned wire.

3.17 Dimensions and weight. The dimension and weight of the cleaner shall not exceed the following:

Width	46 inches
Length	82 inches
Height	66 inches
Dry weight	3,000 pounds

3.18 Protective finish. The protective finish shall be paint. All surfaces to be painted shall, immediately prior to painting, be cleaned, free of all foreign matter, and dried. The protective paint coating shall be at least 2.0 mils (.002 inch) thick. The paint adhesion shall be tested by making two intersecting cuts in the paint to the base metal, two inches in length. Fiberglass tape shall be applied over the two intersecting cuts and then shall be removed without any peeling of the paint. The surface hardness of the paint shall be between the industrial standards of H to HB pencil lead. The color shall be non-glossy forest green and the markings shall be glossy black.

3.19 Operation diagrams. Instruction or operation manuals shall contain diagrams that illustrate all components and their interconnections for each operational system on the steam cleaner. Systems to be illustrated shall include, but not be limited to:

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a. Water flow.

b. Electrical, both component diagram and ladder diagram.

c. Fuel flow.

d. Cleaning compound.

e. Operational and safety controls.

3.20 Lifting, pulling, and tie-down provisions. Each lifting, pulling, and tie-down device shall be identified and marked or stenciled. Lifting and pulling rings or eyes shall be not less than three inches in inside diameter, shall be located at the ends of the skid, and shall maintain clearance between slings and exterior parts of the cleaner, spreader bars considered. The lifting rings or eyes shall support the cleaner and skid loaded to no less than 2-1/2 times the cleaner's wet weight (the weight with all tanks full and all hoses and guns stowed). The lifting, pulling, and tie-down devices shall withstand a horizontal longitudinal load 4 times the wet weight and a horizontal transverse load of no less than 2 times the wet weight. The loads shall be divided equally among the lifting, pulling, or tie-down devices.

3.21 Racks. Racks shall be attached to the cleaner for stowing the cleaning gun assemblies, nozzles, hoses and the power cable when not in use. The racks shall be positioned so that stowed equipment is not exposed to heat from the burner.

3.22 Fungus treatment. The use of fungus supporting nutrient materials shall be avoided whenever possible. If any such materials are used in the cleaner, they shall be treated with varnish conforming to MIL-V-173 using one percent copper 8 - quinolinolate except as noted in a and b below:

a. Components of circuit elements that are inherently fungus and moisture resistant, or which are hermetically sealed, shall not be treated.

b. Components of circuit elements whose functions shall be adversely affected by the varnish shall not be treated.

3.23 Compound metering device. The metering device shall inject liquid cleaning compound into a chosen volume of clear water. The metering device shall be adjustable for rates of from 0 to no less than 10 percent, cleaning compound-to-water, by volume. The metering device shall have an indicator showing the mix ratio in one-half percent increments and marked at one percent intervals. When adjusted to the desired rate, the device shall maintain a constant delivery to within $\pm$ 10 percent of the setting.

3.24 Lubrication. All bearings, except sealed-for-life type, and all other working parts shall be provided with standard means for lubrication. When shipped, all oil reservoirs, grease fittings, and those parts requiring lubrication shall be full of the manufacturer's recommended lubricant.

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3.25 Interchangeability. All replaceable parts shall be manufactured to such standards as shall permit replacement without the need for modification. Threads shall comply with FED-STD-H28 and the applicable Detailed Standard section therein. No testing in FED-STD-H28 shall be required.

3.26 Hourmeter. The cleaner shall be provided with a 50/60 Hz electrically driven hourmeter. The hourmeter shall be a digital counter type and shall record 9,999 operating hours in 1/10-hour increments. The hourmeter shall be mounted on the control panel of the cleaner and connected to the pump motor circuit.

3.27 Information marking.

3.27.1 Identification marking. The cleaner shall be marked for identification in accordance with MIL-STD-130 and shall include the National Stock Number. Identification marking shall also include the words: "Cleaner wired for 208 volts, 50/60 Hz, single phase current."

3.27.2 Control marking. All controls shall be identified for the function performed and for direction of operation for the change desired. Control marking shall be on securely attached name plates or etched or embossed plastic or metal, or an adhesive backed metal (mylar) type plates. Decalcomania is not acceptable.

3.27.3 Instruction panel. An instruction panel which shall be impervious to oil, grease, and weather shall be mounted on the cleaner in an accessible and practical location. The instructions shall be on etched or embossed plastic or metal or on adhesive backed metal (mylar) type plates. Decalcomania is not acceptable. The panel shall set forth the following information:

a. Detailed operating instructions.

b. Lubricating instructions.

c. Instructions for water supply connection.

d. Instructions for blowdown.

3.27.4 Cleaning compound identification. The words "Use only steam cleaning compound conforming to MIL-C-22542" shall be conspicuously marked on the exterior of the cleaner.

3.27.5 Fuel identification. A plate shall be securely attached near the fuel filler stating "Use Only diesel Fuel Grade No. 1 or No. 2." The plate shall be impervious to oil, grease and weather and shall conform to the requirements of 3.27.2.

3.27.6 Discharge information. The rated discharge pressures shall be permanently marked on the exterior of the cleaner.

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3.27.7 Electrical ground identification. The electrical grounding terminal specified in 3.5.5 shall be conspicuously marked.

3.28 Humidity conditions. The cleaner shall not be damaged and shall be capable of meeting the requirements of 3.5.3 through 3.5.3.3 after having been exposed for 24 hours to, and while still at, a relative humidity of 97 to 100 percent and an air temperature of 80 degrees F to 85 degrees F, air movement and radiation negligible.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that the products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item noted in 3.1. First article inspection shall consist of the examination in 4.5 and all tests under 4.7. The tests shall be performed in accordance with the test conditions in 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (k) as listed. Failure of any item to pass an examination or test shall be cause for rejection of the item.

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(a) Product examination (4.5).
(b) Hydrostatic pressure tests (4.7.1).
(c) High voltage requirement test (4.7.2).
(d) Safety controls tests (4.7.3).
(e) Cold water discharge test (4.7.4.1).
(f) Hot water discharge test (4.7.4.2).
(g) Operational test (4.7.4.3.2).
(h) Continuous duty test (4.7.5).
(i) Noise level test (4.7.6.2).
(j) Paint test (4.7.8).
(k) Packaging validation tests (4.7.9).
(l) Packaging inspection (4.8).

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or Test	Lot or Batch size	Sample size
Product examination (see 4.5)	-	100 percent
Hydrostatic pressure test (see 4.7.1)	-	100 percent
High voltage requirement test (see 4.7.2)	1-150	5
Safety controls test (see 4.7.3)	-	100 percent
Cold water discharge test (see 4.7.4.1)	-	100 percent
Hot water discharge test (see 4.7.4.2)	-	100 percent
Operational test (see 4.7.4.3.2)	-	100 percent
Continuous duty test (see 4.7.5)	-	1/
Noise level test (see 4.7.6.2)	-	2/
Paint test (see 4.7.8)	1-150	3
Packaging validation tests (see 4.7.9)	-	3/
Packaging inspection (see 4.8)	1-150	3

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100 percent inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

1/ The continuous duty test shall be conducted on one of the first three cleaners produced if a first article test is not required by the contract. If a first article test is required, the continuous duty test shall be performed as part of the first article test. In either case, additional cleaners shall be subjected to the continuous duty test as a portion of every comparison test required by the contract.

2/ The noise level test shall be conducted even if a first article test is not required by the contract. The test shall be performed on the first production cleaner only.

3/ The packaging validation tests shall be conducted on one of the packaged steam cleaners selected from the first lot submitted for quality conformance inspection and tests.

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4.5 Product examination. Visually, manually, and dimensionally examine the cleaner to determine conformance with the requirements of 3.2 through 3.2.3 and 3.3.7 through 3.28. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of moveable parts by hand to assure proper functioning. Dimensional examinations shall include overall dimensions (see 3.17) and other dimensions and weights as specified. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the cleaner and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement parts will degrade the form, fit, or function of the end item (see 3.25), shall be cause for rejection.

4.6 Test conditions. The cleaner shall be tested and inspected under the listed conditions:

4.6.1 Preparation for test. The cleaner shall be prepared for test by servicing, lubricating, adjusting, and running-in as specified by the operator's manual. All controls, valves, and pump(s) shall be checked to assure proper operation.

4.6.2 Measurements.

4.6.2.1 Pressure. Pressure shall be measured at the inlet and outlet sides of the heating coil. Pressure shall be measured using calibrated equipment capable of providing accuracy within 1 percent over the full scale of the gage.

4.6.2.2 Temperature. Temperature measurements shall be taken at the inlet and outlet sides of the heating coil. Temperature shall be measured using calibrated equipment capable of providing accuracy within 1 percent over the full scale of the gage.

4.6.2.3 Fuel consumption. Fuel usage shall be determined by the weight loss method, and the period between weighings of the fuel shall represent at least five minutes of burner operation. Fuel weight shall be measured on a calibrated scale or balance having an accuracy within 2 percent of the measured value, and reading in pounds.

4.6.2.4 Liquid flow. Liquid flow rates shall be determined by use of a flow meter which shall be of the totalizing and recording type and accurate within 3 percent of the measured value. Readings shall be in gallons per time period. Instantaneous flowmeter readings shall not be used.

4.6.3 Fuel. The cleaner shall be supplied with No. 2 diesel fuel for all tests specified herein.

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4.6.4 Water supply. The water supply for the tests shall be sufficient to supply the cleaner with a flow of at least 600 gallons per hour. Water under pressure of up to 125 psig may be used for all tests except the suction supply water source test (see 4.7.6.1). The temperature of the water supplied to the cleaner shall be as specified for the individual tests.

4.6.5 Chemical additives. Chemical additives such as cleaning solution or descaling solution shall be used in the cleaner only when specified for the individual tests.

4.7 Tests. Cleaners shall be tested and inspected under the test conditions specified in 4.6 and all subparagraphs thereof.

4.7.1 Hydrostatic pressure tests.

4.7.1.1 Heating coil. The fabricated heating coil assembly shall be pressure tested prior to assembly into the cleaner. Fill the coil with water and apply hydrostatic pressure of no less than 2000 psig. Block the inlet and observe for a period of 10 minutes. Pressure drop in excess of 10 psig resulting from leakage shall be cause for rejection (see 3.7).

4.7.1.2 Water piping. Fill all water piping components between the pump and nozzle with water, block the flow at the gun nozzle, and apply hydrostatic pressure of no less than 1050 psig. Block inlet and observe for a period of 10 minutes. Pressure drop in excess of 10 psig resulting from leakage shall be cause for rejection (see 3.7).

4.7.2 High voltage requirement test. A test voltage of at least 1416 volts rms, 60 Hz, shall be applied between each and every isolated circuit in the unit. The voltage shall be supplied by a transformer having a volt-ampere capacity of 500 VA without a voltmeter, but if less than 500 VA, the test equipment shall have a voltmeter or other indicator in the output circuit to assure the test voltage does not fall below 1416 volts during the tests. The voltage shall be applied to an uninsulated point in each circuit and between each circuit and frame or chassis ground. The voltage shall be maintained for at least 60 seconds. When test voltage is applied to the power input terminals the master switch and other involved switches shall be in the closed position. During this part of the test solid state components and electric motors may be damaged. This damage is not cause for rejection but all damaged solid state components and electric motors shall be replaced or repaired upon conclusion of the test. When testing internal circuits, solid state devices and electric motors may be bridged or removed to prevent damage. Any damage to any part other than solid state components and electric motors evidenced by malfunction, breakdown, arcing, or other surface or air discharge shall be cause for rejection (see 3.5.5). CAUTION: Due to high voltage, extreme care should be taken during this test.

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4.7.3 Safety controls tests. The cleaner safety devices shall be tested as specified and listed. All safety controls tests shall be conducted with the cleaner operating under 60 Hz power (see 3.5).

4.7.3.1 Power loss test. With the cleaner operating in the cold water mode, the cleaner electrical power shall be momentarily interrupted. The cleaner electrical power shall then be interrupted in the same manner while the cleaner is operating in the hot water mode and in the vapor discharge mode. Automatic restart of any function of the cleaner after a power interruption in any mode (without resetting the master switch) shall be cause for rejection (see 3.5.1).

4.7.3.2 Burner malfunction test. The cleaner shall be subjected to each of the following conditions while set up for operation in the hot water mode, and in the vapor discharge mode:

(a) The continuous spark ignition device shall be displaced from the fuel flow so that the device will continue to operate without causing ignition. An attempt shall then be made to start up and operate the cleaner in the normal manner. The ignition device shall then be returned to its normal position, and the cleaner started.

(b) With the cleaner operating with a normal flame, the air inlet for the burner shall be blocked so that a flameout occurs. An attempt shall be made to start up and operate the cleaner in the normal manner. The blockage shall then be removed and the cleaner re-started.

Failure of the cleaner to automatically stop all flow of fuel to the burner within forty five seconds of the occurrence of either of the above conditions or the ability to re-start the cleaner without activating a manual reset shall be cause for rejection (see 3.5.2).

4.7.3.3 Inadequate water flow test. The cleaner overtemperature and overpressure safety devices shall be disconnected or otherwise defeated, and the cleaner subjected to each of the following conditions while operating in the hot water mode and in the vapor mode:

(a) The amount of water supplied to the cleaner shall gradually be reduced to zero, such that the cleaner is allowed to draw air and pump itself dry. The water supply to the cleaner shall then be restored. Failure of the cleaner to automatically stop all flow of fuel to the burner before the water flow rate falls below 150 gph or failure of the burner to automatically re-start when the water supply is restored shall be cause for rejection (see 3.5.2).

(b) The flow of water at the pump inlet shall be gradually restricted until the pump inlet is completely blocked. The blockage shall then be removed. Failure of the cleaner to automatically stop all flow of fuel to the burner before the water flow falls below 150 gph or failure of the burner to automatically re-start when the blockage is removed shall be cause for rejection (see 3.5.2).

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(c) The flow of water at the coil inlet shall be gradually restricted until the coil inlet is completely blocked. The blockage shall then be removed. Failure of the pump bypass system to bypass the entire pump output before the water pressure exceeds 575 psig in the hot water mode or 375 psig in the steam mode; or failure of the cleaner to automatically stop all flow of fuel to the burner before the water flow rate falls below 150 gph; or failure of the burner to automatically re-start when the blockage is removed shall be cause for rejection (see 3.5.2 and 3.5.3).

NOTE: Following this test, all cleaner safety devices shall be returned to their original state.

4.7.3.4 Temperature limit device malfunction test. With the cleaner operating in the hot water mode, the device(s) controlling the water temperature shall be defeated or adjusted so as the water temperature increases, fuel flow to the burner shall automatically stop before the discharge temperature reaches 260 degrees F. If the fuel flow is automatically terminated, the cleaner shall be allowed to cool down until the burner automatically re-starts. The cleaner shall then be operated in the vapor mode, and the device(s) controlling the vapor temperature defeated or adjusted so as the vapor temperature increases, fuel flow to the burner shall automatically stop before the discharge temperature reaches 360 degrees F. If the fuel flow is automatically terminated, the cleaner shall be allowed to cool down until the burner automatically re-starts. Failure of the cleaner to automatically stop all flow of fuel to the burner before the water temperature exceeds 260 detrees F in the hot water mode and before the vapor temperature exceeds 360 degrees F in the vapor mode; failure of the cleaner to maintain the flow of water through the coil after fuel flow has been terminated; or failure of the burner to automatically re-start after the temperature falls back within acceptable limits shall be cause for rejection (see 3.5.2). NOTE: If necessary, the cleaner pressure regulating device(s) may be bypassed or taken out of the circuit for the duration of this test. At the conclusion of this test all safety devices shall be returned to their original state.

4.7.3.5 Pressure limit device malfunction test. With the cleaner operating in the hot water mode and the valves on the discharge guns closed so that the pump is operating in full bypass, the pressure controlling device(s) for that mode shall be adjusted or defeated so that the pressure at the discharge manifold is raised until either the burner ceases operation or the pressure exceeds 600 psig. If the burner automatically ceases operation, it shall be allowed to cool down until the pressure drops to within the operating limits. The cleaner shall then be operated in the vapor mode with the valves on the discharge gun closed, and the pressure regulating device(s) for that mode shall be adjusted or defeated so that the pressure at the discharge manifold is raised until either the burner ceases operation or the pressure exceeds 400 psig. If the cleaner automatically ceases operation, it shall be allowed to cool until the pressure drops to within the operating limits. Failure of the burner to automatically and completely cease operating before the pressure exceeds 600 psig in the hot water mode and 400 psig in the vapor mode or failure of the cleaner to automatically re-start after the pressure

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falls back within the operating limits shall be cause for rejection (see 3.5.2). NOTE: If necessary, the cleaner temperature regulating and flow sensing devices may be defeated or removed from the circuit for the duration of this test. Following this test, all safety devices shall be returned to their original state.

4.7.4 Performance tests.

4.7.4.1 Cold water discharge test. The cleaner shall be operated in the cold water mode for at least 15 minutes under 60 Hz power and for at least 15 minutes under 50 Hz power. For the first article test only, the temperature of the inlet water shall be less than 50 degrees F for at least 5 minutes of both the 60 Hz and 50 Hz portions of the test. Failure of the cleaner to continuously discharge at least 500 gph of water through the rinse gun at a pressure of at least 400 psig for the full period of operation under either source of power shall be cause for rejection (see 3.3.1.4).

4.7.4.2 Hot water discharge test. The cleaner shall be operated in the hot water mode under 60 Hz power for 15 minutes after the water temperature at the coil outlet has stabilized and for the same period under 50 Hz power. During the first article test only, the temperature of the inlet water shall be 40 degrees F or less for the duration of both the 60 Hz and 50 Hz portions of the test. Failure of the cleaner to continuously discharge at least 500 gph of water heated to a temperature of not less than 160 degrees F at a pressure of at least 400 psig for 15 minutes while operating under either source of power shall be cause for rejection (see 3.3.1.3).

4.7.4.3 Vapor discharge tests.

4.7.4.3.1 Cold start test. The cleaner shall be configured for operation under 50 Hz power using No. 2 Diesel fuel. Operate in the cold water mode for 5 minutes to insure that the coil is cool. The cleaner shall then be set for operation in the vapor mode, supplied with water at a temperature of not more than 40 degrees F, and operated for eight minutes. Failure of the cleaner to discharge vapor heated to a temperature of at least 300 degrees F at a pressure of at least 200 psig and a rate of at least 600 gph by the end of the eight minute period shall be cause for rejection (see 3.3.1.1). On first article test, check thermal efficiency at this point (see 4.7.5.2).

4.7.4.3.2 Operational test. The cleaner shall be operated in the vapor mode under 50 Hz power and under 60 Hz power. The cleaner shall be operated for 15 minutes under each source of power after the temperature at the coil outlet has stabilized. During each 15 minute operating period, the pressures at the pump outlet and at the coil outlet shall be measured and recorded. Failure of the cleaner to continuously discharge water vapor heated to a temperature of at least 300 degrees F at a pressure of at least 200 psig and at a rate of at least 600 gph for the full 15 minute period while operating under each source of power shall be cause for rejection (see 3.3.1.1 (a)). NOTE: When this test is done as a part of first article test the cleaner shall also be tested for suitability of operation with appropriate small nozzle in conformance with 3.3.1.1(b), and suitability of operation with two guns in conformance with 3.3.1.1(c). Vapor discharge continuity shall be verified by

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switching from two gun operation at 300 gph per gun to one gun operation at 300 gph in conformance with 3.3.1.2. Then change from the vapor discharge mode to the hot water discharge mode and verify that the pressure change is accomplished without requiring a pressure regulator or similar device to be adjusted.

4.7.5 Continuous duty test. The cleaner shall be operated in the vapor mode for a period of 400 hours. The cleaner shall then be operated in the hot water mode for 48 hours. This 448 hour test shall be performed for no less than 8 hours per calendar day and shall be completed within 56 consecutive calendar days (see 3.3.2). Water hardness shall be no less than 12 grains per gallon (see 3.7.3 and 3.4). During these tests, data shall be recorded as specified in 4.7.5.1.

During the continuous duty test, testing shall be as follows:

(a) During the vapor mode the cleaner shall be operated in each of the five attitudes for a period of 80 consecutive hours. These are horizontal and tipped 15 degrees to the front, rear, and to each side. Verify cleaner operation and that there is no leakage from the tanks during the tilted tests (see 3.3.3).

(b) Verify the cleaning solution tank has a capacity of at least 10 gallons and operate cleaner for at least 4 hours using the cleaning solution (see 3.10).

(c) Verify the fuel tank has sufficient capacity to operate the cleaner for a minimum of 4 hours in the vapor mode (see 3.9).

(d) Conduct the thermal efficiency test during the first 5 hours, at 200 plus or minus 5 hours, and within 5 hours of the end of the test. Verify efficiency is no less than 80 percent at any time (see.3.3.1.5).

(e) Conduct a smoke density test at 195 plus or minus 5 hours and at 395 plus or minus 5 hours. Verify the density is no greater than a number 2 smoke spot (see 3.3.2).

(f) Measure temperature on all valves, controls, and component surfaces contacted by the operator and verify that none exceed 130 degrees F (see 3.5).

(g) After completion of test verify that the heating coil can be easily replaced without damage to the coil, coil jacket, or support, and can be removed using only common hand tools and by one person (see 3.7). Replace with a new coil.

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During the continuous duty test, servicing shall be as follows:

(a) Blowdown frequency shall not exceed one per 8 consecutive operating hours (see 3.4.a and 3.7.3).

(b) Chemical descaling frequency shall not exceed one per 80 operating hours (see 3.4.b and 3.7.3).

(c) Desooting shall not be required during the 448 hour test (see 3.4.c).

Any breakdown or replacement of major components which include pumps, motors, burners, safety controls and devices, heating coil, and transformers, or failure to meet requirements of 4.7.5 (a) through (g) and (a) through (c), shall be cause for rejection (see 3.3.2).

4.7.5.1 Data readings. During the continuous duty test specified in 4.7.5, the following performance data shall be recorded as a minimum at intervals of not more than one operating hour, except where shorter intervals are required to determine compliance with specified requirements. The maintenance information specified below shall be recorded each time the cleaner is maintained, serviced, or adjusted during the continuous duty test.

4.7.5.1.1 Performance data.

(a) Date.
(b) Time.
(c) Hour meter reading (cleaner).
(d) Inlet pressure at coil.
(e) Outlet pressure at coil.
(f) Liquid flow rate in gph (water or solution).
(g) Feed water temperature entering cleaner.
(h) Liquid temperature leaving cleaner (water or solution at coil outlet)
(i) Ambient temperature.
(j) Voltage.
(k) Amperage.
(l) Fuel consumption in pounds per minute and BTU rating (only required when determining thermal efficiency).
(m) Motor and/or pump speeds (rpm).
(n) Cleaner serial number.

4.7.5.1.2 Maintenance information.

(a) Date, time, and cleaner hourmeter reading at which the action was initiated.
(b) Type of maintenance, servicing, or adjustments performed and manhours to accomplish.
(c) Date, time, and cleaner hourmeter reading when the action was completed.

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4.7.5.1.3 Corrective action. Failure of a cleaner to meet the continuous duty test requirement shall be cause for deferring further submission of cleaners for Government acceptance until the contractor has completed the following actions:

(a) Conduct a failure analysis study, performing a dimensional, physical, and visual examination, as required, of the component(s) or condition(s) suspected to be the cause of failure.

(b) Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect in future production.

(c) Examine cleaners awaiting submittal for Government acceptance, partially assembled cleaners, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect(s) is purged from the inventory and not presented to the Government for acceptance.

(d) Provide objective evidence of the results of the failure analysis and corrective actions to the Government representative prior to submitting the reconditioned test sets for retest.

4.7.5.2 Thermal efficiency test. With the cleaner operating in the vapor mode and the coil outlet temperature stabilized, measurements of fuel consumption, water flow, and coil inlet and outlet temperatures during a period of at least 15 minutes of operation shall be recorded; and the cleaner thermal efficiency calculated as:

Calculation - Thermal Efficiency:

$$E = \frac{Qw\ (T)\ (C)\ (D)}{Qf\ (H)} \times 100$$

Where:

Qw = water flowing through the coil (gal).
C = specific heat of water (1.0 BTU/lbm F).
D = 8.33 lbm/gal.
Qf = fuel consumed (lbm).
H = energy content of fuel (BTU/lbm).
T = difference between coil inlet and outlet water temperatures (F).

Failure of the cleaner to demonstrate a thermal efficiency of at least 80 percent shall be cause for rejection (see 3.3.1.5). The thermal efficiency 200 hours test shall be conducted at 60 Hz as part of the continuous duty test (see 4.7.5). The 50 Hz thermal efficiency test shall be conducted on the first article test only, after the cold start test (see 4.7.4.3.1).

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4.7.5.3 Blowdown test. After the cleaner has been in operation a sufficient amount of time to build up scale deposits, as demonstrated by a decrease in the coil outlet pressure, a blowdown operation shall be performed in accordance with the manufacturer's instructions. Failure of the blowdown operation to return the coil outlet pressure to normal, or evidence of damage to any component of the cleaner as a result of the blowdown operation shall be cause for rejection (see 3.7.3). This test is to be performed as a part of the continuous duty test (see 4.7.5).

4.7.6 Environmental and function tests.

4.7.6.1 Suction supply water source. With the 25 foot suction hose, foot valve, and suction strainer attached, the cleaner shall be operated under 50 Hz power in the cold water mode to draw water against a resistance equivalent to a 20 foot vertical rise. This may be accomplished either by drawing from a standing body of water whose surface is at least 20 feet lower in altitude than the water pump inlet, or by drawing against an equivalent vacuum of 20 feet at standard atmospheric pressure at sea level. Failure of the cleaner to draw the water and discharge it at a rate of at least 500 gph and a pressure of at least 400 psig shall be cause for rejection (see 3.6.4 and 3.3.1.3).

4.7.6.2 Noise level tests. With the cleaner in operation in the vapor mode with the burner on, noise level readings shall be taken with a sound level meter meeting Type 1 requirements of American National Standard Specification for sound level meters, ANSI S1.4, set on the slow response of the A scale. Cleaner noise level measurements shall be taken at a distance of no greater than 10 feet from the perimeter of the cleaner at a height of five feet with no effect on the measurements from the cleaner gun and nozzle. Noise levels measured in excess of 85 dbA shall be cause for rejection (see 3.5.6).

4.7.6.3 Cleaning solution metering test. Set metering valve at 10. Operate the cleaner in the hot water mode at a flow rate of 500 to 510 gph. Record the time to use one gallon of cleaning solution and record the total gun output in gallons for the same period of time. Set the metering valve at 5 and repeat the test. Failure of the metering valve to provide cleaning solution to water ratio at a constant delivery rate within plus/minus 10 percent of the meter device setting shall be cause for rejection (see 3.23).

4.7.6.4 Altitude test. The cleaner shall be placed in a test chamber or tested under conditions to maintain a pressure equivalent to a 6,000-foot altitude and a maximum ambient temperature of 107 plus 5 minus 0 degrees F. It shall then be subjected to the tests in 4.7.4.2 and 4.7.4.3.2. Scientific computation (stoichiometric calculations based on fuel analysis and velocity pressure in the combustion chamber) may be submitted in lieu of conducting this test. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2 under the above conditions shall be cause for rejection (see 3.3.4).

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4.7.6.5 High operating temperature test. The cleaner, after being in an ambient air temperature of 120 degrees F to 125 degrees F until stabilized, and while at that temperature, shall be tested in accordance with 4.7.4.2 and 4.7.4.3.2. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2 while in the above condition shall be cause for rejection (see 3.3.5).

4.7.6.6 Low operating temperature test. The cleaner, after being in an ambient air temperature of 35 degrees F to 40 degrees F until stabilized, and while at that temperature, shall be tested in accordance with 4.7.4.2 and 4.7.4.3.2. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2 while in the above condition shall be cause for rejection (see 3.3.5).

4.7.6.7 High storage temperature test. The cleaner and accessories shall be submitted to a 24 hour cycle of temperature starting with ten hours at 80 plus or minus 5 degrees F, followed by five hours of steady increase to 150 plus or minus 5 degrees F, four hours at temperature, and five hours of steady decrease to 80 plus or minus 5 degrees F. The equipment shall then be inspected for evidence of damage and tested in accordance with 4.7.4.2 and 4.7.4.3.2. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2, or evidence of damage to any cleaner component after exposure to the above conditions shall be cause for rejection (see 3.3.6).

4.7.6.8 Low storage temperature test. The cleaner and accessories shall be placed in an ambient air temperature of -50 plus or minus 5 degrees F for a period of 72 hours. At the end of this time, the equipment shall be removed and allowed to stabilize at an air temperature of 80 plus or minus 5 degrees F. The equipment shall then be inspected for evidence of damage and tested in accordance with 4.7.4.2 and 4.7.4.3.2. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2, or evidence of damage to any cleaner component after exposure to the above conditions shall be cause for rejection (see 3.3.6).

4.7.6.9 Humidity test. The cleaner and accessories shall be subjected to a 24 hour exposure of 97 to 100 percent relative humidity at 80 degrees to 85 degrees F air temperature, air movement and radiation effects negligible. The cleaner shall then be tested in accordance with 4.7.4.2 and 4.7.4.3.2. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2, or evidence of damage to any cleaner component after exposure to the above conditions shall be cause for rejection (see 3.28).

4.7.7 Transportability tests. Any visible permanent deformation or breakage of the lifting, pulling, or tie-down devices; or visible damage to the cleaner, its components, frame, or skid as a result of any of the tests specified shall be cause for rejection (see 3.3.7).

4.7.7.1 Lifting rings or eyes (Vertical loading). Sufficient weight shall be attached to the cleaner to bring the total weight of the cleaner plus the added weight to a minimum of 2.5 times the wet weight (the weight of the cleaner with all tanks full and all hoses and guns stowed). The cleaner shall then be suspended from its four lifting eyes or rings at a height of approximately one foot above the ground for a minimum of 30 seconds (see 3.20).

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4.7.7.2 Lifting, pulling, and tie-down devices (Horizontal loading). The lifting, pulling, and tie-down devices shall be subjected to the listed horizontal load tests (see 3.20).

4.7.7.2.1 Horizontal longitudinal load. A load equal to 4 times the cleaner's wet weight shall be equally divided among the lifting, pulling or tie-down devices and applied in each horizontal longitudinal direction for a minimum of 30 seconds.

4.7.7.2.2 Horizontal transverse load. A load equal to 2 times the wet weight of the cleaner shall be equally divided among the lifting, pulling or tie-down devices and applied in each horizontal transverse direction for a minimum of 30 seconds.

4.7.7.3 Fork lift test. The cleaner shall have all hoses and gun mounted. The water and fuel tanks shall be full (wet weight). The cleaner shall then be lifted from each of its four sides by using a fork lift truck without fork extensions. The cleaner shall be raised to a height of no less than one foot each time (see 3.3.7).

4.7.7.4 Trailer transport test. Mount the cleaner on the M105, 1.5 ton cargo trailer. Tow the trailer for a distance of 13 miles consisting of 5 miles of primary (paved) road at speeds up to 50 mph, 5 miles of secondary (gravel) road at speeds up to 35 mph, and 3 miles rough terrain (unimproved) at speeds up to 5 mph. Damage to or displacement of any component, accessory, part or tool, or evidence of defective, welded, bolted, or riveted connection, or inability of the cleaner to function properly, shall be cuase for rejection (see 3.3.7). To determine if cleaner is functioning properly, start and operate in the vapor mode for 5 minutes.

4.7.8 Paint test. The hardness, adhesion, and thickness of the paint used on the machine (see 3.18), shall be checked as follows:

(a) The graphite point of a sharpened HB lead pencil shall be pressed at a right angle into a painted area of the unit until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of the paint, shall be cause for rejection.

(b) A test sample of the same metal, painted with the same process and at the same time as the unit, shall have a sharpened H lead pencil pressed into the paint as in (a). Failure of the surface to pit (paint too hard) shall be cause for rejection.

(c) Two, two-inch intersecting lines shall be cut through the paint to the base metal of the test sample. A strip of 1/2 inch wide fiberglass tape shall be firmly pressed over the intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection.

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(d) Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least 0.002 inch thick shall be cause for rejection.

4.7.9 Packaging validation tests. After the Quality Assurance inspection and tests have been successfully completed, one steam cleaner, packaged in accordance with the requirements of Section 5 Packaging, shall be selected at random and subjected to the following rough handling tests specified in FED-STD-101. If a First Article Test (FAT) is required by the contract, the packaging validation test shall be conducted on the steam cleaner used for the FAT. The apparatus used shall be as described in FED-STD-101. A dummy load shall not be used. All tests are to be conducted under ambient conditions. After the test has been completed, the container shall be inspected. Defects in accordance with those described below shall be cause for rejection. The unit shall then be unpacked and again subjected to the Quality Assurance inspections and tests of 4.5 and 4.7.4.3.2 to reaffirm that the steam cleaner packaging meets the requirements specified in Section 5. After all requirements have been met, the steam cleaner shall be inspected and repackaged (see 4.8) before submission for government acceptance. After testing, any damage to the packaging shall be replaced prior to acceptance of the lot.

4.7.9.1 Superimposed-load (stackability) test. Test shall be conducted using Method 5016.1 of FED-STD-101. Criteria for passing/failing of the test is stated in Paragraph 7.1.5 of Method 5016.1.

4.7.9.2 Mechanical handling test. Test shall be conducted using Method 5011.1 Paragraphs 6.2 (Lifting and transportation by forklift truck), 6.3 (Hoisting with slings), and 6.5 (Pushing) of FED-STD-101. Criteria for passing/failing of the test is stated in Paragraph 7.1.5 of Method 5011.1.

4.8 Packaging inspection. The items shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing and marking requirements of Section 5.

5 PACKAGING

5.1 Disassembly. When practical, for the protection of components, attachments and accessories against damage, pilferage or to reduce cubage, items may be disassembled only to a necessary basic degree. Disassembled parts shall be preserved, anchored, braced, blocked and cushioned to prevent damage. Disassembled parts shall be clearly and legibly marked as to identity and proper location on the assembled item. All fasteners removed during disassembly shall be secured in one of the mating parts. A part shall not be removed from an assembly unless it can be reassembled readily in the field without special skills or tools.

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5.2 Disassembled parts marking. Disassembled parts shall be marked, when necessary with instructions, to facilitate reassembly. Removed parts and mating parts on the item shall be marked by stenciling, on the part, or by use of tags. Tags shall be waterproof and ink used (on parts or tags) shall be waterproof. Tags shall be firmly secured to each individual part.

5.3 Preservation. Preservation shall be level A or C, as specified (see 6.2.1).

5.3.1 Level A.

5.3.1.1 Cleaning and drying. The cleaner components and accessories, and all tanks, tubing, piping, fittings, and valves shall be drained, cleaned, dried, and treated with preservative prior to submission for acceptance. All procedures shall be in accordance with the applicable procedures of MIL-P-116.

5.3.1.2 Preservation application. Immediately after cleaning and drying, apply a coat of type P-2 preservative of MIL-P-116 to all exterior, critical, bare, ferrous metal surfaces of the cleaner. All surfaces coated with P-2 shall be covered with barrier material conforming to MIL-B-121 and secured with tape conforming to PPP-T-60. Exterior, noncritical, bare, ferrous metal surfaces, which have not been painted or plated, shall be coated with type P-1 or type P-19 preservative. Interior surfaces of the delivery gun(s) shall be preserved with type P-10, grade 30, preservative oil. Interior surfaces of the fuel system shall be completely flushed (lines and internal filters) and completely fogged (tank and filter(s)) with type P-10, grade 30, preservative oil. Interior surfaces of the water system shall be completely flushed with a solution consisting of 50 percent antifreeze (ethylene glycol) conforming to A-A-870 and 50 percent clean fresh water. Induce sufficient air pressure into the system to remove residual water.

5.3.1.3 Unit protection. The basic cleaner shall be unit packaged in accordance with method I of MIL-P-116. Small openings in the cleaner shall be sealed with tape or plastic cap-plugs conforming to PPP-T-60 or MIL-C-5501, respectively. Large apertures shall be covered with barrier material conforming to PPP-B-1055 (class E-2). Secure barrier material in place with above mentioned tape. Components of the cleaner and accessories shall be provided unit protection as specified in paragraphs 5.2.1.3.1 through 5.2.1.3.8 below.

5.3.1.3.1 Gages and meters. Gages and meters shall be covered with one inch thick uncompressed cushioning material conforming to PPP-C-1797, type I or PPP-C-850, type I or II, class 1 or 2, grade R. Secure cushioning in place with PPP-T-60 tape. When gages and meters are protected by an integral cover, cushioning material will not be required.

5.3.1.3.2 Drive belts. Drive belts shall be either removed or released from tension. The grooves of all ferrous metal pulleys shall be coated with primer conforming to TT-P-664. Belts removed from the cleaner shall be packaged in accordance with method III of MIL-P-116.

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5.3.1.3.3 Electric motors. Electric motors shall be packaged in accordance with level A requirements of MIL-E-16298 and the applicable requirements for assembled machines.

5.3.1.3.4 Switch boxes. Electric switch boxes shall be completely sealed with PPP-T-60 tape.

5.3.1.3.5 Power cable and delivery hoses. The ends of the hoses and power cable shall be wrapped with barrier material conforming to MIL-B-22191, type III. Secure material with PPP-T-60 tape by encircling the hose or cable, as applicable, making a water-tight closure. Individually, coil the hose and cable to a minimum safe diameter and secure coil with a minimum of three (3) ties of cotton tape or twine to prevent uncoiling. Alternatively, the hoses and cable may be coiled on their respective stowage racks and secured to prevent loosening.

5.3.1.3.6 Tools and delivery nozzles. Maintenance tools and delivery nozzles shall be wrapped with barrier material conforming to MIL-B-22191, type III or L-P-378, type III, secured with tape, and placed in their respective stowage compartments.

5.3.1.3.7 Delivery guns. Delivery guns shall be placed and secured in the stowage device provided on the cleaner. Apertures shall be sealed with tape conforming to PPP-T-60, type IV.

5.3.1.3.8 Publications. Technical publications issued with the cleaners shall be packaged in accordance with method IC-1 or MIL-P-116.

5.3.1.3.9 Consolidated packaging. All packaged accessories, technical publications, and disassembled parts, which have not been placed in stowage racks or applicable compartments of the cleaner, shall be adequately cushioned with material conforming to A-A-1898, type II and packaged in one or more snug-fitting fiberboard boxes conforming to PPP-B-636, class weather resistant, type CF. The container shall be waterproof sealed, as specified in the appendix to the box specification.

5.3.2 Level C. Level C preservation shall be as specified in MIL-STD-1190.

5.4 Packing. Packing shall be military level A or C, as specified (see 6.2.1).

5.4.1 Level A. Each cleaner shall be packed in an open crate conforming to MIL-C-104, Type II, Class 2, Style A. The consolidated package shall be placed inside the shipping container and all items secured to restrict any movement. Arrangements of contents, anchoring, blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186.

5.4.2 Level C. Level C packing shall be in accordance with MIL-STD-1190.

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5.5 Marking. Marking of both level A and C interior packages and exterior shipping containers shall be as specified in MIL-STD-129 and any special marking, as required therein. Bar code marking is also required and shall be in accordance with MIL-STD-129.

5.6 Rough handling and stackability requirements. The completed unit pack, shipping container(s) and all contents shall be capable of withstanding the shipping and handling conditions expected from fork lift handling and truck transport and stackable a minimum of two each high. The container, its contents and the packaging shall not be damaged, functionally or physically, or items displaced by the appropriate rough handling test(s) specified elsewhere in the contract.

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The steam cleaner covered by this specification is used for cleaning dirt, grease, and heavy oil deposits from transportation vehicles, construction equipment, and miscellaneous items for maintenance and overhaul. The steam cleaner is used in fixed shops as well as in the field.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
(c) First article when required (see 3.1 and 4.3).
(d) Measuring and indicating device graduation(s) required (see 3.2.6.1).
(e) Level of preservation and packing required (see 5.2 and 5.3).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Standard lubricants. The cleaner manufacturer should specify and use to the maximum extent possible those lubricants (see 3.2.3) that are presently in the Department of Defense supply system. MIL-HDBK-113, MIL-HDBK-267, and MIL-STD-838 may be used as selection guides.

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6.6 Subject term (key word) listing.

Blowdown
Vapor mode
Descaling
Flame control
Coil
Burner

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

Army - AL

Review activity:

Army - SM

User activity:

Army - ME

Preparing activity:

Army - AL

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